approach to determine the value of the string coupling constant itself. But, as discussed, perturbative methods are sensible only if the coupling constant is less than 1, and hence our calculation may be making an unjustified assumption about its own answer—namely, that the result will be smaller than 1. Our failure could well indicate that this assumption is wrong and that, perhaps, the coupling in any one of the five string theories is greater than 1. Third, the unwanted flexibility may merely be due to our use of approximate rather than exact equations. For instance, even though the coupling constant in a given string theory might be less than 1, the equations of the theory may still depend sensitively on the contributions from all diagrams. That is, the accumulated small refinements from diagrams with ever more loops might be essential for modifying the approximate equations—which admit many solutions—into exact equations that are far more restrictive.

By the early 1990s, the latter two possibilities made it clear to most string theorists that complete reliance on the perturbative framework was standing squarely in the way of progress. The next breakthrough, most everyone in the field agreed, would require a nonperturbative approach—an approach that was not shackled to approximate calculational techniques and could therefore reach well beyond the limitations of the perturbative framework. As of 1994, finding such a means seemed like a pipe dream. Sometimes, though, pipe dreams spill over into reality.

Duality

Hundreds of string theorists from around the world gather together annually for a conference devoted to recapping the past year’s results and assessing the relative merit of various possible research directions. Depending on the state of progress in a given year, one can usually predict the level of interest and excitement among the participants. In the mid- 1980s, the heyday of the first superstring revolution, the meetings were filled with unrestrained euphoria. Physicists had widespread hope that they would shortly understand string theory completely, and that they would reveal it to be the ultimate theory of the universe. In retrospect this was naive. The intervening years have shown that there are many deep and subtle aspects of string theory that will undoubtedly take prolonged and dedicated effort to understand.